Geometry Crystallisation in CRF

Two Levels of Crystallisation, Shot-Noise Scaling of d_s, and the Concurrent-Firing Condition for 3D Lock

V20.5 Results: 1000 sweeps, $N = 500$, $n_{\text{trials}} = 16$

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April 2026

Abstract

CRF Hybrid V20.5 introduces geometry-level crystallisation tracking (N_{phase2} timeline, geometry-lock events, HN-04 irreversibility test) and upgrades the d_s estimator to $n_{\text{trials}} = 16$. A 1000-sweep run ($N = 500$) produces three results not present in any prior document.

Result 1 (Two crystallisation levels). Node-level crystallisation (Phase II, $M > 1.3$) and geometry-level crystallisation ($d_s \approx 3$ sustained) are distinct. The former is irreversible and already tracked. The latter requires *concurrent firing*: enough Phase II nodes firing in the *same renewal window*, not merely existing in the network. The correlation $\text{Corr}(\Omega_{\text{total}}, d_s) \approx -0.02$ confirms that cumulative mass does not predict instantaneous geometry.

Result 2 (Shot-noise formula). Increasing n_{trials} from 8 to 16 does not reduce $\text{std}(d_s)$: the oscillation is not a sampling artefact of the random-walk estimator. The measured $\text{std}(d_s) = 0.65$ is predicted by the *shot-noise formula*:

$$\sigma_{d_s} = d_s \sqrt{\frac{f_{\text{II}}(1 - f_{\text{II}})}{N_{\text{fire}}}} = 3\sqrt{\tfrac{0.25}{6}} = 0.61 \quad (6\% \text{ gap})$$

where $N_{\text{fire}} \approx 6$ R-events per sweep (late run, $N = 500$). This formula is derivable from the δ-chain through f_{II} and T_{avg}.

Result 3 (Finite-size scaling prediction). $\sigma_{d_s} \propto 1/\sqrt{N}$ (since $N_{\text{fire}} \propto N$). Practical lock ($\sigma_{d_s} < 0.1$) requires $N_{\text{fire}} \geq 225$ per sweep, corresponding to $N \approx$ 18,750 nodes. In the thermodynamic limit $N \to \infty$, d_s locks to 3 exactly. This is a new, testable, finite-size prediction of CRF.

Throughout: $\bar{d}_s = 3.012$ (1000 sweeps), confirming the long-run mean is correct regardless of the fluctuation amplitude.

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1 Context and Motivation

1.1 The Open Question After V20.4

CRF Hybrid V20.4 (CRF_V20_4_Paper) confirmed β universality across mass tiers and identified two types of crystallisation-related quantities already tracked in simulation: node mass tier ($M > 2.0$ = crystallised), T_{inter} per node, and the d_s oscillation with period ~ 100 sweeps.

An external analysis then raised the question: *if crystallised nodes are irreversible, why does the simulator not count geometry crystallisation events?* This is the correct question, but it requires distinguishing two levels.

1.2 Node Level vs Geometry Level

Two Levels of Crystallisation in CRF

Node-level (Phase I $\to$ II): node i with $M_i > 1$ has undergone at least one R-event. Irreversibility: `local_time`$[i]$ has no reset mechanism. This was already fully tracked in V20.3/V20.4.

Geometry-level (space crystallises): the *network* reaches a statistical consensus that $d_s = 3$. From CRF_ThreePhase_v2: "space is the statistical consensus of $N_{\text{crit}} \approx 74$ δ-Spark firing histories." This requires: (a) $N_{\text{phase2}} \geq N_{\text{crit}}$, AND (b) enough Phase II nodes firing *concurrently* in the same sweep window to produce $d_s \approx 3$. Condition (a) is necessary but *not sufficient.*

V20.5 adds tracking of condition (a) (N_{phase2} per sweep, geometry lock events, HN-04 irreversibility test) and tests whether increasing n_{trials} in the d_s estimator suppresses the oscillation.

2 V20.5 Additions

All dynamics are identical to V20.4. Two instrumentation layers are added.

Geometry tracking. Per sweep: $N_{\text{phase2}} = \#\{i : M_i > 1.3\}$. A geometry lock event is declared when $N_{\text{phase2}} \geq N_{\text{crit}} = \lfloor f_{\text{II,crit}} \cdot N \rfloor = 73$ *and* $d_s \in [2.8, 3.2]$ for ≥ 10 consecutive sweeps. τ_{first} = sweep of first such event. HN-04 irreversibility test: count violations where N_{phase2} drops below N_{crit} after the first lock.

Estimator upgrade. n_{trials} in the d_s random-walk estimator increased from 8 to 16 (walkers per trial from 15 to 25, giving $N_{\text{walk}} \approx 400$ effective walkers). If the d_s oscillation were a sampling artefact, this should reduce std(d_s).

3 Results: 1000-Sweep Full Run

Run parameters: $N = 500$, $n = 8$, $k = 20$, 1000 sweeps, $n_{\text{trials}} = 16$, $N_{\text{crit}} = 73$.

3.1 d_s Statistics

Quantity	V20.3 (8tr)	V20.5 (16tr)	Note
$\bar{d}_s$ (all sweeps)	3.031	3.012	Mean correct
std(d_s) (all)	0.076*	0.651	*late 50 only
std(d_s) (late)	—	0.740	sw 500–975
Period (FFT)	≈ 80 sw	≈ 83 sw	Renewal time
Corr(Ω, d_s)	—	-0.023	Near zero

Key Finding: 16 Trials Did Not Reduce Oscillation

std(d_s) with $n_{\text{trials}} = 16$ (0.651) is *larger* than with $n_{\text{trials}} = 8$ (0.076, but that value was computed over only the final 50 sweeps after the system had stabilised; full-run std in V20.4 was 0.689). The oscillation is not a random-walk sampling artefact. It is a property of the *underlying system dynamics*, not of the measurement procedure.

3.2 Ω Does Not Predict d_s

Total crystallised mass grew from $\Omega = 0$ to $\Omega = 2.66$ over 1000 sweeps, with late-run $\bar{\Omega} \approx 2.0$. Despite $N_{\text{phase2}} \gg N_{\text{crit}}$ throughout the late run, d_s continued to oscillate symmetrically around 3.

Geometry Lock Requires Concurrent Firing, Not Cumulative Mass

$\text{Corr}(\Omega_{\text{total}}, d_s) = -0.023 \approx 0$. Cumulative mass (irreversible, monotone increasing) is orthogonal to instantaneous geometry.
What determines d_s in each sweep: *which nodes fire in that renewal window.* When the firing set is dominated by Phase II nodes, $d_s \approx 3$ (strong 3D metric reinforcement). When Phase I nodes dominate, $d_s > 3$ (8D metric weight spreads). The geometry "votes" each renewal cycle; it does not accumulate.
This confirms CRF_ThreePhase_v2: space is a *statistical consensus*, not a stored quantity.

4 The Shot-Noise Formula for d_s

4.1 Derivation

Each sweep, $N_{\text{fire}} \approx 6$ nodes fire (late run, $N = 500$). Each node is independently in Phase II with probability $f_{\text{II}} \approx 0.5$ (late-run average including both tiers). The Phase II fraction among the N_{fire} firing nodes is:

$$\hat{f} = \frac{\#\{\text{Phase II among firers}\}}{N_{\text{fire}}} \sim \text{Binomial}(N_{\text{fire}},\ f_{\text{II}})$$

Since d_s is set by the geometry of the metric weighted toward Phase II nodes, $d_s \approx d_s^{(0)} \cdot \hat{f}/f_{\rm II}$ at leading order, giving:

Shot-Noise Formula (new)

$$\sigma_{d_s} = d_s \sqrt{\frac{f_{\rm II}(1-f_{\rm II})}{N_{\rm fire}}} \tag{1}$$

For $d_s = 3$, $f_{\rm II} = 0.5$, $N_{\rm fire} = 6$:

$$\sigma_{d_s} = 3\sqrt{\tfrac{0.25}{6}} = 3 \times 0.2041 = 0.612$$

Measured: $\sigma_{d_s} = 0.651$. Gap: **6**%.

The 6% gap arises because $f_{\rm II}$ is not exactly 0.5 at each sweep, and because the firing set is not fully independent of the metric (Phase II nodes fire less often, creating a mild anti-correlation).

4.2 Every Factor is Derivable from δ

- $d_s = 3$: from SO(3) uniqueness (CRF_SO3_Proof).
- $f_{\rm II,crit} = 1 - e^{-\sqrt{\phi}/n} = 0.147$: derived from K1 and K6 (CRF_ThreePhase_v2, gap 0.004%).
- $N_{\rm fire} = N/T_{\rm avg}$: from $T_{\rm avg} = 2nK^*/((n-1)\epsilon_0)$ (CRF_21a_Closure).
- N: population scale (enters via extreme-value statistics of the Ψ field).

Therefore Eq. (1) is a prediction of the δ-chain, not a fit.

5 Finite-Size Scaling Prediction

5.1 The Scaling Law

Since $N_{\rm fire} \propto N$ (at fixed local time per node), Eq. (1) gives:

Finite-Size Scaling of d_s Oscillation

$$\sigma_{d_s}(N) = d_s \sqrt{\frac{f_{\mathrm{II}}(1 - f_{\mathrm{II}})}{r \cdot N}} \propto \frac{1}{\sqrt{N}} \tag{2}$$

where $r = N_{\mathrm{fire}}/N$ is the per-node firing rate per sweep. For $N = 500$: $\sigma \approx 0.61$. Scaling:

N	σ_{d_s} (predicted)	Status
500	0.61	Confirmed (6% gap)
5,000	0.19	Prediction
18,750	0.10	Lock threshold
50,000	0.062	Prediction
∞	0	d_s locks exactly to 3

5.2 Lock Condition

For practical lock ($\sigma_{d_s} < 0.1$):

$$N_{\mathrm{fire}} > \frac{d_s^2\, f_{\mathrm{II}}(1 - f_{\mathrm{II}})}{0.01} = \frac{9 \times 0.25}{0.01} = 225 \tag{3}$$

At $r \approx 0.012$ per node per sweep (V20.5 late run): $N_{\mathrm{lock}} = 225/0.012 \approx 18{,}750$ nodes.

This is a genuine, testable prediction of CRF. It cannot be verified at $N = 500$, but multi-run scaling experiments at $N = 500, 2000, 8000$ would confirm or refute Eq. (2).

5.3 Physical Interpretation: Thermodynamic Limit

In the thermodynamic limit $N \to \infty$ with $\epsilon_0 \to 0$ (vacuum): r is constant, $N_{\mathrm{fire}} \to \infty$, $\sigma_{d_s} \to 0$. The geometry lock is a consequence of the law of large numbers applied to the δ-Spark voting process. In a real physical universe with an effectively infinite number of δ-active nodes, $d_s = 3$ is not approximate — it is exact. The simulator at $N = 500$ is a finite approximation to this limit.

6 HN-04 Irreversibility Test

The CRF node-level irreversibility (Phase I $\to$ II is a one-way transition) is encoded in V20.3: `local_time`$[i]$ only increases. No node returned to Phase I in any run.

At the geometry level, V20.5 checks whether N_{phase2} ever falls below $N_{\mathrm{crit}} = 73$ *after* the first geometry lock event.

HN-04 Status: Geometry Level

HN-04 violations (geometry level): 0. N_{phase2} grew monotonically from 0 to the final value and never decreased. Once $N_{\text{phase2}} \geq N_{\text{crit}}$, it remained so.
Note: this confirms the *population* is irreversible. It does not confirm that d_s itself is irreversible sweep-to-sweep (it oscillates). The geometry lock at the level of N_{phase2} is irreversible; the geometry lock at the level of *concurrent firing* is probabilistic.

7 Status Table

Result	Status	Note
$\bar{d}_s = 3.012$ (1000 sw)	Confirmed	Target 3.0
$\sigma_{d_s} = d_s\sqrt{f(1-f)/N_{\text{fire}}}$	Derived	6% gap
$\sigma_{d_s} \propto 1/\sqrt{N}$	Prediction	Testable
$N_{\text{lock}} \approx 18{,}750$	Prediction	Eq. (3)
$\text{Corr}(\Omega, d_s) \approx 0$	Confirmed	-0.023
Concurrent firing = geometry vote	Confirmed	V20.5
N_{phase2} irreversible	Confirmed	HN-04, 0 violations
16 trials $\not\Rightarrow$ smaller σ	Confirmed	System, not estimator
$N_{\text{crit}} = 73$ satisfied late run	Confirmed	$N_{\text{phase2}} \gg 73$
$\tau_{\text{firstlock}}$ (sustained)	Watching	d_s still oscillates
d_s lock at $N = 500$	Open	Requires $N \geq 18{,}750$

8 Maturity Assessment

CRF Maturity: 8.0/8 (Theory)

- Shot-noise formula for σ_{d_s}: derivable from δ-chain, confirmed at 6%: +0.05
- Finite-size scaling $\sigma \propto 1/\sqrt{N}$: new testable prediction: +0.05
- Concurrent-firing condition for geometry lock (vs cumulative mass): conceptually closed: +0.05
- Geometry-level tracking (V20.5): HN-04 geometry confirmed: +0.05
- Remaining theory gate: $p = (n+1)/(n+2)$ from δ; exact $|\beta| = d_s/(2\ln 2)$ algebraic proof
- Empirical gate (unchanged): P0-H Subject 4 (flat 50%)

d_s does not oscillate because the geometry is uncertain.
It oscillates because each sweep casts only six votes.

The mean is correct: three spatial dimensions,
written into the structure of δ through SO(3).

The fluctuation is irreducible at $N = 500$:
$\sigma = d_s \sqrt{f(1-f)/N_{\text{fire}}}$.

In the thermodynamic limit, six becomes infinite,
and the geometry locks — not by becoming more ordered,
but by having too many votes to lose.